JEE Mains 2019 Chapter wise Question Bank

Quadratic Equation – Questions

Q1

Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to:

(1) -256 (2) 512

(3) -512 (4) 256

9 Jan Morning

Q2

If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval $[1, 5]$, then m lies in the interval:

(1) $(-5, -4)$ (2) $(4, 5)$

(3) $(5, 6)$ (4) $(3, 4)$

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Q3

The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is:

(1) 3 (2) 2

(3) 4 (4) 5

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Q4

If 5, 5r, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to:

(1) $\frac{3}{4}$ (2) $\frac{5}{4}$

(3) $\frac{7}{4}$ (4) $\frac{3}{2}$

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Q5

Consider the quadratic equation

$(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval $(0, 2)$ and its other root lies in the interval $(2, 3)$. Then the number of elements in S is:

(1) 18 (2) 12 (3) 10 (4) 11

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Q6

The value of λ such that sum of the squares of the roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is:

(1) $\frac{15}{8}$ (2) 1

(3) $\frac{4}{9}$ (4) 2

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Q7

If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube of the other root, then a value of k is :

(1) -81 (2) 100

(3) 144 (4) -300

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Q8

If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is :

(1) $2 - \sqrt{3}$ (2) $4 - 3\sqrt{2}$

(3) $-2 + \sqrt{2}$ (4) $4 - 2\sqrt{3}$

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Q9

The number of integral values of m for which the quadratic expression, $(1+2m)x^2 - 2(1+3m)x + 4(1+m), x \in R$, is always positive, is :

(1) 3 (2) 8

(3) 7 (4) 6

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Q10

If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is :

(1) 2 (2) 5 (3) 4 (4) 3

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Q11

The sum of the solutions of the equation $\left|\sqrt{x}-2\right| + \sqrt{x}\left(\sqrt{x}-4\right) + 2 = 0$, $(x > 0)$ is equal to:

(1) 9 (2) 12 (3) 4 (4) 10

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Q12

The number of integral values of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :

(1) 1 (2) 2

(3) infinitely many (4) 3

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Q13

Let p, q ∈ R. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then:

(1) $p^2 - 4q + 12 = 0$ (2) $q^2 - 4p - 16 = 0$

(3) $q^2 + 4p + 14 = 0$ (4) $p^2 - 4q - 12 = 0$

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Q14

If m is chosen in the quadratic equation

$$(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$$

such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is:

(1) $10\sqrt{5}$ (2) $8\sqrt{3}$ (3) $8\sqrt{5}$ (4) $4\sqrt{3}$

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Q15

If α and β are the roots of the quadratic equation, $x^2 + x \sin\theta - 2\sin\theta = 0\ 0, \theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12}+\beta^{12}}{(\alpha^{-12}+\beta^{-12})(\alpha-\beta)^{24}}$ is equal to :

(1) $\frac{2^{12}}{(\sin\theta-4)^{12}}$ (2) $\frac{2^{12}}{(\sin\theta+8)^{12}}$

(3) $\frac{2^{12}}{(\sin\theta-8)^{6}}$ (4) $\frac{2^{6}}{(\sin\theta+8)^{12}}$

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Q16

The number of real roots of the equation $5 + \left|2^x - 1\right| = 2^x(2^x - 2)$ is:

(1) 3 (2) 2 (3) 4 (4) 1

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Quadratic Equation - Solutions

Q1

(1) Consider the equation

$$x^2 + 2x + 2 = 0$$

$$x = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$$

Let $\alpha = -1 + i, \beta = -1 - i$

$$\alpha^{15} + \beta^{15} = (-1+i)^{15} + (-1-i)^{15}$$

$$= \left(\sqrt{2}e^{i\frac{3\pi}{4}}\right)^{15} + \left(\sqrt{2}e^{-i\frac{3\pi}{4}}\right)^{15}$$

$$= \left(\sqrt{2}\right)^{15}\left[e^{\frac{i45\pi}{4}} + e^{\frac{-i45\pi}{4}}\right]$$

$$= \left(\sqrt{2}\right)^{15}.2\cos\frac{45\pi}{4} = \left(\sqrt{2}\right)^{15}.2\cos\frac{3\pi}{4}$$

$$= = \frac{-2}{\sqrt{2}}\left(\sqrt{2}\right)^{15}$$

$$= -2\left(\sqrt{2}\right)^{14} = -256$$

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Q2

(2) Given quadratic equation is: $x^2 - mx + 4 = 0$

Both the roots are real and distinct.

So, discriminant $B^2 - 4AC > 0$.

$\therefore \quad m^2 - 4 \cdot 1 \cdot 4 > 0$

$\therefore \quad (m-4)(m+4) > 0$

$\therefore \quad m \in (-\infty, -4) \cup (4, \infty)$ ...(i)

Since, both roots lies in [1, 5]

$\therefore \quad -\frac{-m}{2} \in (1, 5)$

$\Rightarrow \quad m \in (2, 10)$ ...(ii)

And $1 \cdot (1 - m + 4) > 0 \Rightarrow m < 5$

$\therefore \quad m \in (-\infty, 5)$ ...(iii)

And $1 \cdot (25 - 5m + 4) > 0 \Rightarrow m < \frac{29}{5}$

$\therefore \quad m \in \left(-\infty \frac{29}{5}\right)$ ...(iv)

From (i), (ii), (iii) and (iv), $m \in (4, 5)$

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Q3

(1) The roots of $6x^2 - 11x + \alpha = 0$ are rational numbers.

$\therefore$ Discriminant D must be perfect square number.

$$D = (-11)^2 - 4 \cdot 6 \cdot \alpha$$

$= 121 - 24\alpha$ must be a perfect square

Hence, possible values for α are

$$\alpha = 3, 4, 5.$$

$\therefore$ 3 positive integral values are possible.

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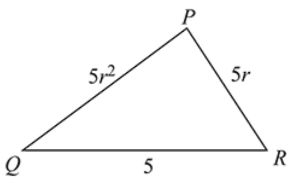
Q4

(3)



ΔPQR is possible if

$$5 + 5r > 5r^2$$

$$\Rightarrow \ 1 + r > r^2$$

$$\Rightarrow \ r^2 - r - 1 < 0$$

$$\Rightarrow \ \left(r - \frac{1}{2} + \frac{\sqrt{5}}{2}\right)\left(r - \frac{1}{2} - \frac{\sqrt{5}}{2}\right) < 0$$

$$\Rightarrow \ r \in \left(\frac{-\sqrt{5}+1}{2}, \frac{\sqrt{5}+1}{2}\right)$$

$$\because \ \frac{7}{4} \notin \left(\frac{-\sqrt{5}+1}{2}, \frac{\sqrt{5}+1}{2}\right)$$

$$\therefore \ r \neq \frac{7}{4}$$

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Q5

(4) Consider the quadratic equation

$$(c-5)x^2 - 2cx + (c-4) = 0$$

Now,

$$f(0).f(3) > 0 \text{ and } f(0).f(2) < 0$$

$$\Rightarrow \ (c-4)(4c-49) > 0 \text{ and } (c-4)(c-24) < 0$$

$$\Rightarrow \ c \in (-\infty, 4) \cup \left(\frac{49}{4}, \infty\right) \text{ and } c \in (4, 24)$$

$$\Rightarrow \ c \in \left(\frac{49}{4}, 24\right)$$

Integral values in the interval $\left(\frac{49}{4}, 24\right)$ are 13, 14, ..., 23.

$$\therefore \ S = \{13, 14, ..., 23\}$$

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Q6

(4) The given quadratic equation is

$$x^2 + (3-\lambda)x + 2 = \lambda$$

Sum of roots $= \alpha + \beta = \lambda - 3$

Product of roots $= \alpha\beta = 2 - \lambda$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (\lambda - 3)^2 - 2(2 - \lambda)$$

$$= \lambda^2 - 4\lambda + 5$$

$$= (\lambda - 2)^2 + 1$$

For least $(\alpha^2 + \beta^2)$ $\lambda = 2$.

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Q7

(4) Let α and β be the roots of the equation,

$$81x^2 + kx + 256 = 0$$

Given $(\alpha)^{\frac{1}{3}} = \beta$

$$\alpha = \beta^3$$

$\because$ Product of the roots $= \frac{256}{81}$

$$\therefore \quad (\alpha)(\beta) = \frac{256}{81}$$

$$\Rightarrow \quad \beta^4 = \left(\frac{4}{3}\right)^4 \quad \Rightarrow \quad \beta = \frac{4}{3}$$

$$\Rightarrow \quad \alpha = \frac{64}{27}$$

$\because$ Sum of the roots $= -\frac{k}{81}$

$$\therefore \quad \alpha + \beta = -\frac{k}{81} \quad \Rightarrow \quad \frac{4}{3} + \frac{64}{27} = -\frac{k}{81}$$

$$\Rightarrow \quad k = -300$$

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Q8

(2) Let roots of the quadratic equation are α, β.

Given, $\lambda = \frac{\alpha}{\beta}$ and $\lambda + \frac{1}{\lambda} = 1 \Rightarrow \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 1$

$$\frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = 1 \quad ...(1)$$

The quadratic equation is, $3m^2x^2 + m(m-4)x + 2 = 0$

$\therefore \quad \alpha + \beta = \frac{m(4-m)}{3m^2} = \frac{4-m}{3m}$ and $\alpha\beta = \frac{2}{3m^2}$

Put these values in eq (1),

$$\frac{\left(\frac{4-m}{3m}\right)^2}{\frac{2}{3m^2}} = 3$$

$\Rightarrow \quad (m-4)^2 = 18$

$\Rightarrow \quad m = 4 \pm \sqrt{18}$

Therefore, least value is $4 - \sqrt{18} = 4 - 3\sqrt{2}$

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Q9

(3) Let the given quadratic expression

$(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, is positive for all $x \in R$, then

$$1 + 2m > 0 \quad ...(1)$$

$D < 0$

$\Rightarrow \quad 4(1 + 3m)^2 - 4(1 + 2m)4(1 + m) < 0$

$\Rightarrow \quad 1 + 9m^2 + 6m - 4[1 + 2m^2 + 3m] < 0$

$\Rightarrow \quad m^2 - 6m - 3 < 0$

$\Rightarrow \quad m \in (3 - 2\sqrt{3}, 3 + 2\sqrt{3})$

From (1)

$\therefore \quad m > -\frac{1}{2}$

$m \in \left(3 - 2\sqrt{3}, 3 + 2\sqrt{3}\right)$

Then, integral values of $m = \{0, 1, 2, 3, 4, 5, 6\}$

Hence, number of integral values of $m = 7$

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Q10

(3) The given quadratic equation is $x^2 - 2x + 2 = 0$

Then, the roots of the this equation are $\frac{2 \pm \sqrt{-4}}{2} = 1 \pm i$

Now, $\frac{\alpha}{\beta} = \frac{1-i}{1+i} = \frac{(1-i)^2}{1-i^2} = i$

or $\frac{\alpha}{\beta} = \frac{1-i}{1+i} = \frac{(1-i)^2}{1-i^2} = i$

So, $\frac{\alpha}{\beta} = \pm i$

Now, $\left(\frac{\alpha}{\beta}\right)^n = 1 \Rightarrow (\pm i)^n = 1$

$\Rightarrow$ n must be a multiple of 4.

Hence, the required least value of $n = 4$.

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Q11

(4) Let $\sqrt{x} = a$

$\therefore$ given equation will become:

$|a-2| + a(a-4) + 2 = 0$

$\Rightarrow |a-2| + a^2 - 4a + 4 - 2 = 0$

$\Rightarrow |a-2| + (a-2)^2 - 2 = 0$

Let $|a-2| = y$ (Clearly $y \geq 0$)

$\Rightarrow y + y^2 - 2 = 0$

$\Rightarrow y = 1$ or -2 (rejected)

$\Rightarrow |a-2| = 1 \Rightarrow a = 1, 3$

When $\sqrt{x} = 1 \Rightarrow x = 1$

When $\sqrt{x} = 3 \Rightarrow x = 9$

Hence, the required sum of solutions of the equation $= 10$

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Q12

(3) Given equation is

$(1+m^2)x^2 - 2(1+3m)x + (1+8m) = 0$

$\because$ equation has no real solution

$\therefore D < 0$

$\Rightarrow 4(1+3m)^2 < 4(1+m^2)(1+8m)$

$\Rightarrow 1 + 9m^2 + 6m < 1 + 8m + m^2 + 8m^3$

$\Rightarrow 8m^3 - 8m^2 + 2m > 0$

$\Rightarrow 2m(4m^2 - 4m + 1) > 0 \quad \Rightarrow 2m(2m-1)^2 > 0$

$\Rightarrow m > 0$ and $m \neq \frac{1}{2} \quad \left[\because \frac{1}{2} \text{ is not an integer}\right]$

$\Rightarrow$ number of integral values of m are infinitely many.

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Q13

(4) Since $2-\sqrt{3}$ is a root of the quadratic equation $x^2 + px + q = 0$

$\therefore 2+\sqrt{3}$ is the other root

$\Rightarrow x^2 + px + q = [x-(2-\sqrt{3})][x-(2+\sqrt{3})]$

$= x^2 - (2+\sqrt{3})x - (2-\sqrt{3})x + (2^2 - (\sqrt{3})^2)$

$= x^2 - 4x + 1$

Now, by comparing $p = -4, q = 1$

$\Rightarrow p^2 - 4q - 12 = 16 - 4 - 12 = 0$

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Q14

(3) Sum of roots $= \dfrac{3}{m^2+1}$

$\because$ sum of roots is greatest. $\therefore m = 0$

Hence equation becomes $x^2 - 3x + 1 = 0$

Now, $\alpha + \beta = 3, \ \alpha\beta = 1 \Rightarrow |-\alpha-\beta| = \sqrt{5}$

$|\alpha^3 - \beta^3| = |(\alpha-\beta)(\alpha^2+\beta^2+\alpha\beta)| = \sqrt{5}(9-1) = 8\sqrt{5}$

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Q15

(2) Given equation is, $x^2 + x\sin\theta - 2\sin\theta = 0$

$\alpha + \beta = -\sin\theta$ and $\alpha\beta = -2\sin\theta$

$$\frac{(\alpha^{12}+\beta^{12})\alpha^{12}\beta^{12}}{(\alpha^{12}+\beta^{12})(\alpha-\beta)^{24}} = \frac{(\alpha\beta)^{12}}{(\alpha-\beta)^{24}}$$

$$\therefore |\alpha-\beta| = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{\sin^2\theta + 8\sin\theta}$$

$$\therefore \frac{(\alpha\beta)^{12}}{(\alpha-\beta)^{24}} = \frac{(2\sin\theta)^{12}}{\sin^{12}\theta(\sin\theta+8)^{12}} = \frac{2^{12}}{(\sin\theta+8)^{12}}$$

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Q16

(4) Let $2^x - 1 = t$

$5 + |t| = (t+1)(t-1) \Rightarrow |t| = t^2 - 6$

When $t > 0$, $t^2 - t - 6 = 0 \Rightarrow t = 3$ or -2

$t = -2$ (rejected)

When $t < 0$, $t^2 + t - 6 = 0 \Rightarrow t = -3$ or 2 (both rejected)

$\therefore 2^x - 1 = 3 \Rightarrow 2^x = 4 \Rightarrow x = 2$

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